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The impact of financial innovation, economic complexity, and green technologies on natural resource management in eu countries

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Abstract

The importance of sustainable natural resource management, which seeks to optimize the development and utilization of resources, has grown due to the pressures of economic growth and the depletion of natural resources. The sustainability of natural resources remains a key focus for researchers and policymakers, who are developing effective policies to manage natural resources sustainably, thereby ensuring long-term growth and development. The literature provides limited findings concerning the factors that directly affect natural resource management. This study examines the impact of economic complexity, financial development, financial innovation, a circular economy, and green technological innovation on natural resource management in EU countries from 2004 to 2023. The study uses panel EGLS, Cup-FM, Ba-OLS, Emirmahmutoglu, and Kose causality analysis to identify the objectives. The findings suggest that economic complexity and financial development contribute to the conservation of natural resources in EU countries. In contrast, a circular economy, financial innovation, and green technological innovation negatively affect natural resource conservation. Therefore, it is essential to implement policies to avoid the rebound effect of technological progress in these countries. The causality findings confirm the existence of bidirectional causal relationships between economic complexity, financial development, financial innovation, the circular economy, green technology, and natural resource management. Therefore, there is a need for interactive policies to achieve natural resource management and sustainable development goals within this group of countries.

Keywords: Natural resource management, Circular economy, Financial innovation, Green innovation, Sustainable development, EU countries

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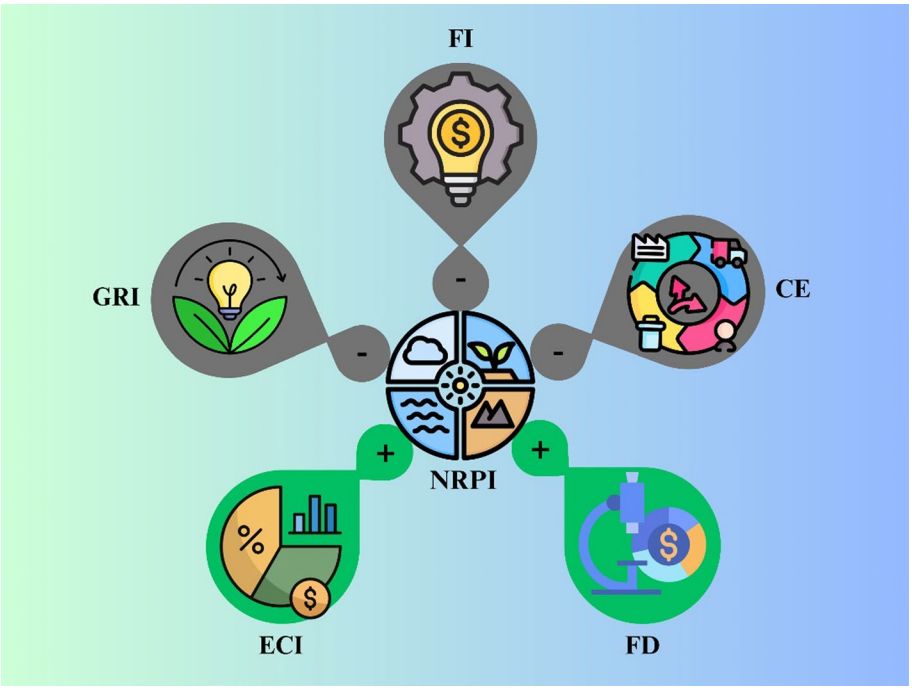
Graphical abstract



Introduction

Individuals and organizations continue to consume large quantities of natural resources, the main drivers of economic development, as if they will always remain abundant (IISD 2021; George et al. 2015). At this consumption rate, material use is expected to increase by more than 60% from 2020 to 2060 (Global Resources Outlook 2024). Although a significant portion of financing is allocated to protecting natural resources and the environment, the identified investment gap still requires additional funding (European Investment Bank 2022). Financial innovations can provide the necessary funds to protect natural resources and, thus, the environment by diversifying financing alternatives (Chishti and Sinha 2022). Today, innovative financial products designed to protect the environment have undergone significant evolution. These include biodiversity credits, climate resilience bonds, conservation impact bonds, environmental impact bonds, green bonds, green securitization, green loans, payments for ecosystem services, rainwater credit trading schemes, sustainability bonds, and sustainability-linked bonds, among others (IISD and NAP 2024). However, the EU financial system heavily depends on banks offering traditional financing instruments (European Commission Directorate 2022).

Moreover, many uncertainties exist regarding the use and management of innovative financial instruments in the EU (Ferrer et al. 2012). Hence, financial innovations have significant potential in protecting natural resources. However, the field is still in its infancy (Almeida et al. 2022) and requires further research.

Energy innovation can contribute to the conservation of natural resources by reducing resource extraction and utilization (Batool et al. 2023a, 2023b). Green technological innovation can enhance the allocation of natural resources and efficient utilization (Miao et al. 2017; Li 2023). Technological innovation can facilitate a transition from the utilization of exhaustible to renewable energy resources (Ali et al. 2023). Using renewable energy can prevent the depletion of natural resources (Lu et al. 2023). Therefore, converting to green energy and increasing energy efficiency can help conserve natural resources in the EU (Andersen et al. 2024). However, the energy efficiency gains accompanying technological advances can create a rebound effect when they are used to produce more or make products cheaper, resulting in increased resource use. Thus, technological progress may nullify gains when accompanied by additional resource consumption (Rosenbaum 2019; Lange et al. 2023; Midden et al. 2007). As a result, while technical developments might support resource conservation and efficient use, they can also exacerbate natural resource management due to rebound effects.

Economic complexity, which is an instrument of the ability to produce sophisticated products (Hidalgo 2021; Hausmann and Hidalgo 2013), can improve resource utilization through technological innovations (Hidalgo and Hausmann 2009; Li et al. 2024a, 2024b; Canh et al. 2020; Dogan et al. 2021). Enhancing innovative skills and knowledge can promote economic diversification and lessen dependence on resources for economic growth (Hausmann and Hidalgo 2011). EC may increase R&D activities, thereby encouraging new technological developments. This is linked to structural changes in the economy, which prompt nations to adopt energy efficiency and renewable energy sources, reduce the use of fossil fuels, and promote sustainable growth (Rafei et al. 2022; Ul-Durar et al. 2023). Although increasing economic complexity can reduce resource consumption through energy efficiency, investment in renewable energy sources requires intensive use of critical minerals such as copper, nickel, cobalt, lithium, chromium, zinc, and aluminum (IEA 2022). Therefore, producing sophisticated and high-tech items that arise with increasing economic complexity may lead to the rapid depletion of natural resources. According to Harvard Growth Lab' Country Rankings (2021), 18 EU member states rank in the top 30 of 133 countries in terms of economic complexity. EU countries, which rank high in economic complexity, may lead to increased consumption of critical minerals when producing sophisticated products. Therefore, the impact of EU policies on natural resource conservation remains to be determined.

The circular economy is an appropriate model for achieving "strong sustainability" (Rocha et al. 2023). "Strong sustainability" emphasizes the preservation of natural resources essential for human well-being at all costs (Godin et al. 2022). The circular economy can achieve "strong sustainability" by meeting social needs while reducing energy and material demands (Lefstad 2021). The CE can support the conservation of natural resources only if secondary production activities reduce primary production. Otherwise, secondary and primary production increase resource extraction, and the circular economy rebound effect occurs (Zink and Geyer 2017; Castro et al. 2022). The EU has developed several policies promoting the circular economy, including the Circular Economy Action Plan, the European Green Deal (2020), and the Sustainable Products Initiative (European Commission 2023). All these efforts and policies aim to manage

natural resources effectively. However, the EU recycling rate is only 11.8% (European Commission 2024), indicating a need for policy adjustments.

This study examines the non-obvious impacts of economic complexity, a circular economy, financial innovation, and green technological innovation on natural resource management in the EU. There are many reasons why this study focuses on NRM policies in the EU. According to the WWF (2019), EU countries, which account for only 7% of the world's population, consume approximately 20% of the world's biocapacity. The EU utilizes between 70 and 97% of the "safe operating space" available worldwide and ranks among the most prominent resource consumers globally. The EU consumes 14.8 tons of fossil fuels, biomass, metals, and minerals per person per year, which is more than double the threshold considered equitable and sustainable. As one of the world's largest resource consumers, it is difficult to argue that the EU's resource management policies are successful (European Environmental Bureau 2024). Moreover, the multidimensional nature of natural resource management requires multifaceted policies. However, natural resource management policies in the EU have been isolated from technological progress and socioeconomic aspects such as financial innovations and economic complexity. Therefore, it is imperative to restructure the EU's natural resource management policy framework. This paper aims to develop these policies by addressing the following research questions: (1) Can technological innovation facilitate natural resource management? (2) Can financial innovations based on innovative technologies improve natural resource management? (3) How has the circular economy impacted natural resource management in EU countries? (4) How can economic complexity help natural resource management?

This paper contributes to the current literature by addressing these questions in multiple ways. First, while the literature extensively examines the effect of technological innovations on natural resource management, there remains a need to clarify the impact of financial innovations on natural resources. It has rarely been the subject of research. This study aims to address the gap in the literature by examining the impact of financial innovations on natural resource management. Second, another significant contribution is the examination of the effects of the circular economy. While circular economy activities can enhance natural resource management, the benefits of the circular economy may need to offset the costs incurred. This study examines the effectiveness of circular economy policies in EU countries. It also serves as an example for other countries, revealing the impacts of circular economy policies on natural resource conservation in the EU. Third, the current study aims to inform the literature and policymakers by investigating the unevenly agreed-upon effects of economic complexity —a more comprehensive indicator than economic growth —on natural resources. Fourth, this study reveals the effectiveness of natural resource management policies by examining EU countries that consume significant global natural resource reserves while prioritizing technological innovation.

Although the analysis is valid only for EU countries, it would be correct to say that the results are generalizable, considering that the policies in EU countries set an example for other countries. Finally, to the best of our knowledge, this is the first study to examine the impact of the circular economy and green technological innovation on natural resource management, employing a holistic approach to present the various aspects of financial

innovations and economic complexity. The remaining sections have the following structure. The following section summarizes the literature and the hypotheses that have been developed. Sect. "Variable discussion, model construction, and methodological approach" discusses the variables, model construction, and methodology. Sect. "Results and discussion" provides the findings from the empirical research. Sect. "Conclusion and discussion" concludes the paper. Section develops policy recommendations.

Theoretical underpinnings and empirical literature

In recent years, the significance of natural resource management (NRM) has increased due to the pressures of economic growth on natural resources and their ongoing depletion. It is now a vital instrument for easing resource shortages. Sustainable NRM involves maximizing the efficiency of resource development and use, creating a cycle in which the use of natural resources promotes both economic and social development (Li 2023). The low efficiency of resource use poses severe problems for the global ecosystem. Hence, natural resource sustainability remains a key focus for researchers and policymakers seeking to achieve balance in the world's ecosystems (Tan et al. 2023). Previous studies provide numerous empirical findings on the connection between technological, economic, and social variables and natural resource rents (Canh et al. 2020; Zhang and Khan 2023; Li et al. 2023; Ul-Durar et al. 2023).

However, studies on NRM are limited. Previous studies on natural resources have focused mainly on the environmental impacts of natural resource rents (Redmond and Nasir 2020; Haseeb et al. 2021; Destek et al. 2023; Badeeb et al. 2023). However, prior studies have largely overlooked the multidimensional aspects of NRM. The current study differs from prior studies investigating the complex relationships among economic complexity (EC), financial development (FD), the circular economy (CE), green innovation (GRI), financial innovation (FI), and NRM. In this context, empirical research and theoretical discussions on the impact of the relevant variables on the NRM are presented in the subheadings.

Economic complexity and natural resource management

The connection between economic growth and natural resources remains unclear in the literature. Several studies (Rahim et al. 2021; Zhang et al., 2023) have demonstrated that natural resource abundance negatively impacts economic growth, a phenomenon known as the resource curse. In contrast, some research has shown that abundant natural resources contribute to economic growth (Redmond and Nasir 2020; Haseeb et al. 2021; Badeeb et al. 2023). However, studies on the effect of economic growth on NRM are limited. Large economies pursuing comprehensive growth often rely on traditional natural resources and energy sources, which can lead to faster depletion and environmental degradation. However, sustainable economic growth practices can prevent the exploitation of natural resources (Abbas et al. 2023). Some empirical results on the link between economic growth and NRM are analyzed. Tan et al. (2023) investigated the linkage between economic growth and NRM in Belt and Road countries for the first time, covering the period from 2006 to 2021. They discovered that economic growth has a positive impact on NRM. According to the

authors, this result depends on national growth models, and resource management may be enhanced if growth is dependent on renewable resources. On the other hand, in a sample of emerging nations, Zhang et al. (2024b) discovered an inverse U-shaped association between economic growth and NRM. After a certain point, urbanization, increased industrial activity, and changes in consumption patterns resulting from economic expansion contribute to the adverse effects of GDP per capita growth on preserving natural resources.

However, the economic complexity index (ECI), which gauges the economy's knowledge-based and sophisticated production structure, has recently been linked to environmental quality in numerous studies and utilized as a predictor of economic growth in the literature (Nathaniel 2021; Lee et al. 2022; Balsalobre-Lorente et al. 2024b; Wang et al. 2024b). EC has a strong relationship with trade in green products and technologies. Countries with high technological sophistication are well-positioned to produce and export green products, and are expected to enhance their competitiveness during the transition to a green economy (Mealy and Teytelboym 2022). An increase in EC will also increase the human capital index by increasing a country's collective level of expertise. The development of skills and expertise can support sustainable development by promoting green innovation and development (Wang et al. 2024a). Similarly, Ahmad and Satrovic (2023) explained that the rationale behind EC supporting environmental sustainability is that the knowledge accumulated in the production systems of economies is not only limited to existing production processes but can also pave the way for cleaner and greener technological innovations.

The literature states that EC has two different effects on natural resources. Hence, the need for natural resources is crucial for the expansion of complex economies, which could lead to increased resource demand. Therefore, the level of EC has a direct impact on the environment and the ecosystem, leading to various forms of pollution. On the other hand, EC may increase R&D activities, thereby encouraging new technological developments. This is linked to structural changes in the economy, which prompt nations to adopt energy efficiency and renewable energy sources, reduce the use of fossil fuels, and promote sustainable growth (Rafei et al. 2022; Ul-Durar et al. 2023). Stated differently, by fostering innovation and expanding technological capacity, ECs can help overcome technical barriers to developing renewable energy. This may lead to technological advancements that facilitate the switch to a more sustainable energy system (Wang et al. 2023). Based on the theoretical discussions above, it can be argued that EC can either improve or complicate NRM.

However, according to Olaniyi and Odhiamba (2024), natural resource-rich nations have the potential to use their rents as investments in innovative and productive industries. Such investments support economic diversification and contribute to developing knowledge intensity and technological capabilities, allowing them to produce more complex exports. However, many resource-rich countries face unethical practices, such as political interference, rent-seeking, and corruption, which are associated with weak institutional structures. This prevents the utilization of natural resource revenues in ways that increase ECs, making them dependent on low-innovation and nontechnology-based sectors. As a result, these countries may not fully realize the potential for economic diversification, limiting their economic development. There are many references

in the literature concerning the impact of ECs and natural resource rents on the sustainable environment (Alvarado et al. 2021; Qian and Madni 2022; Abbas et al. 2023; Saud et al. 2023; Zhang et al. 2024a; Li et al. 2024a). However, the relevant literature offers limited evidence on the direct impacts of ECs on natural resource extraction and rent.

Consequently, Canh et al. (2020) reported that EC increased the rent of coal but decreased the rent of mining and natural gas in their global study covering the period from 2007 to 2017. The authors also observed that EC negatively affects natural resource rents and that improving the production system standard can reduce these rents. In contrast, Ul-Durar et al. (2023) reported an inverted U-shaped relationship between ECs and natural resource rents in 20 resource-rich nations from 2000 to 2021. The authors claim that EC benefits NRM.

Additionally, there is currently no research exploring the direct impact of EC on NRM. Rafei et al. (2022) reported that from 1995 to 2017, countries with diverse institutional qualities improved their environmental quality through the interaction of EC and natural resource revenue. The authors argue that utilizing natural resource rents to fund investments, strengthen infrastructure, develop knowledge-based products, and advance research and development (R&D) will enhance environmental quality. From an alternative standpoint, Wang et al. (2023) discovered a positive influence of natural resources and EC on renewable energy in China during the period 1980–2019. The authors emphasized that EC and natural resources are crucial for developing renewable energy. In contrast, Li et al. (2024b) argued that there are causal linkages between ECs and natural resources in 10 newly industrialized countries from 1975 to 2020 and that policies should be formulated for natural resources and ECs together for sustainable development.

On the other hand, Saud et al. (2024) stated that sustainable development is an approach to development that aims to support the efficient management of natural resources, advanced production technologies, renewable energy sources, and economic prosperity sustainably for future generations and reported that EC positively affected sustainable development in EU countries from 1990 to 2019. From a different perspective, Zhang et al. (2024a, b, c) reported that dependence on natural resources led to price volatility and sustainability concerns in G-20 countries from 1990 to 2021, whereas economic complexity caused market fluctuations. Ahmadi et al. (2025) revealed that there is a positive relationship between EC and the green economy in Iran from 2014 to 2021.

Financial development and natural resource management

Numerous scholarly works have analyzed the role of FD in the relationship between natural resources and economic growth, particularly within the framework of the resource-finance-growth nexus. A robust financial system not only channels natural resource revenues to the real sector but also plays a crucial role in facilitating the exploration of natural resources, thereby stimulating economic growth (Erdogan et al. 2020; Ali and Ramakrishnan 2022). FD is critical in optimizing resource allocation to support economic development. Increased FD accelerates the flow of capital, making high-productivity companies more appealing and enhancing the industrial structure. These shifts in industrial composition can spur technological advancement and knowledge innovation, replacing inefficient, high-consumption industrial

patterns and thereby improving natural resource efficiency. However, overdependence on natural resources can weaken the foundation of FDs by reducing demand for financial products, which has adverse effects (Xu and Tan 2020). Numerous studies have explored the financial resource curse, which posits that resource-rich nations tend to have less developed financial systems (Asif et al. 2020; Guan et al. 2020; Tang et al. 2022; Cui et al. 2023; Alsagr 2024). However, studies investigating the influence of FD on natural resources and NRM are limited. Research has focused primarily on the linkage between FD and environmental sustainability, often analyzing natural resources and FD within the same framework (Wang et al. 2020; Zia et al. 2021; Usman et al. 2022; Opuala et al. 2023; Yazici et al. 2025; Achuo et al. 2025).

Green technological innovation and natural resource management

While green technologies support sustainable resource management, developing countries struggle to access them. The GRI is critical for sustainable economic growth. However, while technological advances can improve energy efficiency at the micro level, they can increase energy consumption and environmental degradation at the macro level. Technological advances in the industrial sector can increase production, leading to a greater demand for energy and raw materials, which can negatively impact environmental quality (Ali et al. 2021; Chu 2022; Li 2023; Balsalobre-Lorente et al. 2024a).

Empirical findings investigating the direct impact of the GRI on NRM are scarce. However, the literature provides evidence of the positive effects of GRI on environmental sustainability, which is explained by increased energy efficiency and the efficient use of natural resources (Javed et al. 2023; Nketiah et al. 2024; Sharif et al. 2024). Ikram et al. (2021) examined the linkages between sustainable investments, green technologies, and cleaner production in Pakistan. They reported that the more a particular company is engaged in promoting cleaner production practices, the more willing it is to invest in green technologies, and that green technologies have the most significant impact on the energy and agriculture sectors. Sharif et al. (2023) reported that in Nordic nations, green technology development processes contribute to the green growth of the economy and that promoting the production of green technical equipment, machinery, and engines can help maintain sustainability.

In contrast, Li (2023), who directly investigated the impact of the GRI on NRM, reported that the GRI had a positive effect on NRM in developing Asian countries between 2000 and 2020. On the other hand, Sharif et al. (2024) reported that the GRI contributes to sustainability by improving energy efficiency and consumption in countries with high EFs. In contrast, Si et al. (2024) reported in their research on 262 non-resource-based and resource-based cities in China that the GRI had a nonlinear effect on green economy efficiency between 2003 and 2018. The study revealed that the impact of the GRI on green economic growth is more substantial in resource-based cities than in non-resource-based cities. They also reported that natural resource dependence in resource-based cities creates an exclusion effect that limits human capital and science-technology investments. From a different perspective, Liu et al. (2025) reported that digitalization, foreign direct investment, and economic growth had a positive impact on the development of environment-related technologies in the

G7 countries from 1990 to 2020. Islam (2025) reported that green innovation negatively affected the relationship between economic factors and the SDGs in the G7 countries from 2002 to 2020, emphasizing that this situation reveals the challenges and possible contradictions of sustainable development.

Financial innovation and natural resource management

The rapid development of FI over the last two decades has improved resource allocation efficiency for financial intermediaries and provided new opportunities for green innovation. FI can mobilize the necessary resources for a green transformation by developing new tools for green investment. FI provides intermediaries with more advanced information screening technologies, enabling them to identify and exclude low-quality or fraudulent green projects and lend more to high-quality green innovation projects. In this way, FI can reduce information asymmetry related to green innovation and facilitate its effective development (Yuan et al. 2021; Wang and Li 2024). In this context, while innovation efforts are expected to be directed toward combating climate change, the financialization of these efforts also requires a certain level of innovation. This new channel of financialization is essential because of project risk. In this respect, it can be assumed that FI may indirectly affect environmental degradation (Chisti and Sinha, 2022). Ullah et al. (2024) argued that promoting sustainable FI and green finance can mitigate environmental damage and promote sustainability. Focusing on the connection between digital finance and renewable energy, Yu et al. (2022) argued that digital finance affects household consumption in various ways through innovative products such as mobile payments, online loans, and online insurance, thereby supporting renewable energy by increasing the income level and credit scale of households. Additionally, the literature provides evidence that FI reduces environmental pollution (Feng et al. 2022; Ullah et al. 2024; Huo et al. 2024).

Moreover, FI offers prospects for improving the financial sector and NRM (Muganyi et al. 2022; Lavrinenko et al. 2023; Tiwari et al., 2024). Although the number of studies on FI and NRM is limited in the literature, Tan et al. (2023) reported that the development of FI contributed positively to NRM in Belt and Road countries between 2006 and 2021. The authors stated that FI can contribute to resource management in an economy in various ways, including greening financial services, reducing carbon emissions, and minimizing paperwork. Similarly, Tiwari (2024) confirmed a positive relationship between FI and NRM in India, and Zhang et al. (2024b) confirmed a positive relationship in a sample of developing countries from 2012 to 2022. Tiwari (2024) stated that FI could create a significant transformation in the field of NRM through its ability to increase accessibility to funds, increase transparency in facilitating responsible investment practices and supply chains, and optimize resource use. From a different perspective, Dai et al. (2025) find that financial innovation has a positive effect on economic sustainability in E7 countries from 2000 to 2022, whereas natural resource management has an inverse relationship with economic sustainability. Rahman and Hossain (2025), in a sample of 18 countries from 2013 to 2019, find that financial innovation initially worsens natural resource management. Nevertheless, these adverse effects are reversed under the regulatory influence of good governance. The authors argued that high levels

of financial innovation improve resource allocation by increasing digital financial inclusion and promoting the sustainable use of natural resources.

Circular economy and natural resource management

Innovation is a key driver of economic progress, and the circular economy (CE) is essential for sustainable economic growth. CEs offer a variety of techniques for working with different stakeholders, but these techniques can have different outcomes. Rebound effects are often unexpected and, in the context of CE, are associated with material efficiency rather than energy efficiency. These effects occur when end-of-life products are reintroduced into production and production is scaled up. For example, material consumption per product can increase when recycled materials generate more waste (Bimonte et al. 2021; Castro et al. 2022; Lowe et al. 2024). Many studies have examined the rebound effect (Chen 2021; Zerbino 2022).

On the other hand, Bianchi and Cordella (2023) reported that a CE in EU countries reduces resource extraction. Koval et al. (2023) reported that the CE in Ukraine is a critical aspect of sustainable development policy, particularly with respect to conservation and prosperity. From a different perspective, Khan (2024), presenting a circular ESG approach, argues that integrating ESG and CE approaches will foster innovation and contribute to the debate on implementing and measuring sustainable practices by supporting a just transition to regenerative practices. Bechetti et al. (2025) reported that CE practices can increase the efficiency of investments in renewing natural resources. On the other hand, Xiao (2025) emphasized that circular economy practices are essential for ensuring environmental sustainability in China and that waste management and disposal are the most significant challenges facing the CE.

Research gap

Based on the above discussion, EC and FD have a reciprocal relationship with natural resource consumption. In addition, FI, CE practices, and GRI are widely believed to be critical contributors to NRM. Although empirical evidence sheds light on the linkage between natural resource use and related variables, the debate on NRM remains open for further clarification. This study focuses on examining the consequences of EC and FD, FI, CE, and GRI on NRM in EU countries. The literature needs to provide evidence of the combined impact of these variables on NRM. Uncovering these ambiguous relationships, which are investigated through indirect effects and remain at the theoretical level of discussion, is crucial for ensuring resource efficiency and formulating policy recommendations that promote economic and social development through sustainable use of natural resources.

Variable discussion, model construction, and methodological approach

Variable discussion

The present study examines the factors influencing NRM in EU countries from 2004 to 2023. Undoubtedly, economic activities are related to the utilization of natural resources. The literature provides empirical evidence on economic growth, natural resource use, natural resource rent (Redmond and Nasir 2020; Rahim et al. 2021; Zhang and Khan 2023), and NRM (Tan et al. 2023; Zhang et al. 2024b). However, the direction of these

relationships remains unclear and varies according to the level of development and economic growth models of countries. Although evidence in the literature (Canh et al. 2020; Cui et al. 2023; Alsagr 2024) links EC and FD to natural resource rent, discussions on NRM remain predominantly theoretical and indirect. Again, financial technologies affecting financial services and markets are expected to transform NRM, and a few empirical findings (Tan et al. 2023; Zhang et al. 2024b; Tiwari 2024) support this view.

Moreover, the literature offers empirical evidence that GRI can enhance the use and allocation of natural resources, thereby contributing to NRM (Miao, 2017; Chau et al. 2022; Li 2023; Zhang et al. 2024b). In contrast, research on the direct impact of the CE on NRM, which aims to minimize the consumption of natural resources and is widely recognized for its contribution to NRM and sustainable development, is limited. The literature consists mainly of empirical findings on the connection between the CE and resource extraction (Bianchi and Cordella 2023; Koval et al. 2023). Moreover, the environmental effects of CE and technological innovation are uncertain. Some studies have focused on the rebound effect (Zink and Geyer 2017; Chen 2021; Zerbino 2022; Lowe et al. 2024).

Based on the above discussions, EC, FD, FI, CE, and GRI, which are believed to impact NRM, are included in the model as explanatory variables. Unlike previous studies (Tan et al. 2023; Li 2023; Zhang et al. 2024b), we included the EC, FD, and CE variables in the model. Complex economies rely on natural resources for growth, which can lead to increased demand for these resources. However, it can also stimulate R&D activities and support the development of new technologies (Rafei et al. 2022). The literature provides little evidence of the direct impact of ECs on natural resource consumption, rent, and management. Revenue from natural resources flows into the real sector via the financial system, with FDs contributing to economic growth. However, it is crucial to assess the effects of FD on NRM and economic variables. Research has also explored whether CE techniques support NRM and sustainable development or if they lead to rebound effects. Therefore, investigating these complex interactions is critical for developing new approaches and providing comprehensive policy implications to enhance NRM.

Following the methodology of Tan et al. (2023) and Zhang et al. (2024b), the natural resource protection indicator (NRPI) (CIESIN 2022) is utilized as the dependent variable to assess NRM. The NRPI evaluates a country's commitment to protecting habitats and conserving biodiversity by determining if at least 17% of all biomes (such as deserts, grasslands, forests, water areas, and tundra) are protected (CIESIN 2022). The index was developed via principal component analysis (PCA) to integrate the FI and CE variables. In the empirical literature, Tan et al. (2023), and Zhang et al. (2024b) used fixed broadband and mobile subscriptions (per 100 inhabitants) and internet users (as a percentage of the population) as FI indicators. In addition, Bernier and Plouffe (2019) included R&D expenditure in the financial intermediation sector (current prices in national currency) as an indicator of FI. From a different perspective, in their study on the development of financial technology, Kowalewski and Pisany (2023) used indicators of fintech startups, fintech activity, and fintech financing as measures of fintech. Building on previous studies, the current research constructs an FI index using the indicators of mobile phone subscriptions, fixed broadband subscriptions, internet usage by individuals, and

Table 1 Describing variables and data sources

	Variables	Acronym	Definition	Sources
Dependent	Natural resource protection indicator	NRPI: log of NRPI	17% protection across all biomes on a 0–100 scale	CIESIN
Independent	The financial innovation index is constructed by principal component analysis based on four components:	FI		
	Mobile cellular subscriptions		Per 100 people	WB
	Fixed broadband subscriptions		Per 100 people	WB
	Individuals using the Internet		% of population	WB
	R&D expenditures in the financial intermediation industry		National currency current prices	OECD
Independent	The circular economy index is constructed by principal component analysis based on three components:	CE		
	Patents related to recycling and secondary raw materials		Number	EUROSTAT
	Recycling rate of municipal waste		Percentage	EUROSTAT
	Trade in recyclable raw materials		Total tonne	EUROSTAT
Independent	Economic complexity	ECI	Economic complexity index	OEC
Independent	Green innovation	GRI	Patents on environmental-Total, percentage	OECD
Independent	Financial development	FD: log of FD	Financial development index (Broad-based index of financial depth, access and efficiency)	IMF

R&D expenditures in the financial intermediation sector, for which data are available. To develop a CE index, three indicators, patents related to recycling and secondary raw materials, a recycling rate of municipal waste, and trade in recyclable raw materials, which represent economic, environmental, and social impact dimensions and have the most extended possible time data, have been used (Manea et al. 2021). The variable definitions, data sources, and characteristics are shown in Table 1.

The estimation model of the study consists of eleven different variables. These variables are collected from six different databases (CIESIN (2024), WB (2024), OECD (2024), EUROSTAT (2024), OEC (2024), and IMF (2024)) for EU countries. Since this study aims to reveal the factors affecting natural resource management, the NRPI data have been decisive for its scope. There are currently twenty-seven EU member states. However, in the present study, the NRPI data, which constitute the dependent variable of the research, for Austria, Belgium, the Czech Republic, Germany, Greece, Hungary, Italy, Poland, Portugal, Slovakia, Slovenia, and Spain were found to take the same and equal values in the sample period (2004–2023). In this case, the NRPI data for these countries may result in a near-singular matrix error, which could compromise the results. These countries could not be included in the analysis to achieve consistent, unbiased, and reliable results in panel data estimation. Therefore, the study covers eight EU countries (Bulgaria, Denmark, Finland, France, Lithuania, the Netherlands, Romania, and Sweden) based on the availability of NRPI data. This study spans from 2004 to 2023, utilizing the most extensive dataset available on the variables, which was supplemented

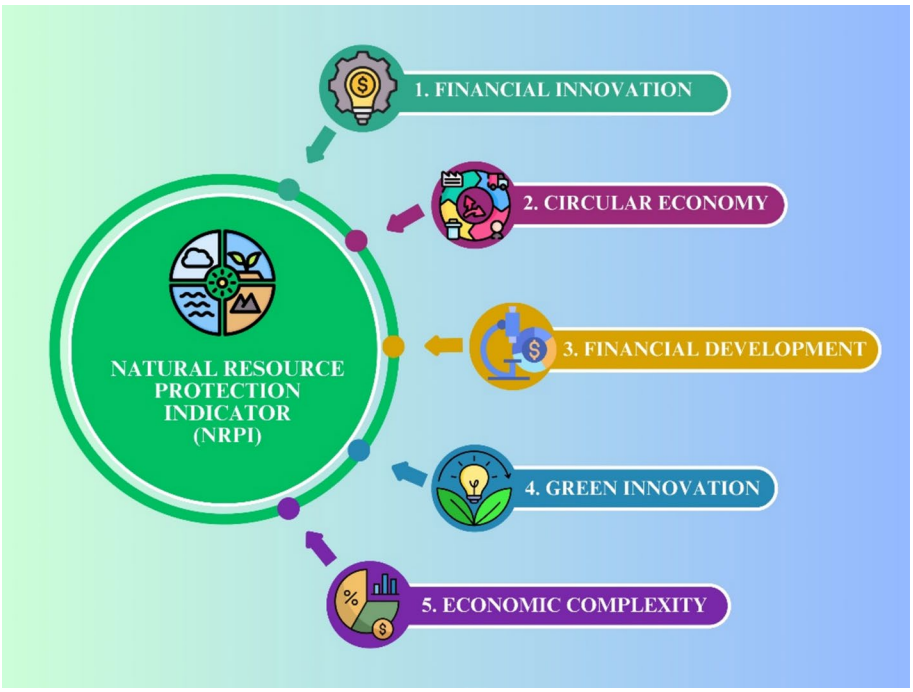


Fig. 1 Framework of the research

by mean imputation. Figure 1 displays the variables analyzed in this study for their effect on NRPI.

Economic modeling

Considering the empirical literature on the connections between NRPI, FI, CE, ECI, GRI, and FD, this study presents a unique research model. The FI and CE indices estimated in the model are constructed. The i and t data are combined to form a panel dataset. The study develops an econometric model to identify the long-run relationships between natural resource management and financial, economic, and technological variables in EU countries. This model employs a panel data analysis to reach groupwise results for the eight EU countries that meet the parameter constraints. The present study examines the factors influencing natural resource management in EU countries as part of this goal. NRM is quantified via NRPI, whereas FI, CE, ECI, GRI, and FD are explanatory variables. The NRM determinants are shown in Eq. (1).

$$NRPI_{it} = f(FI_{it}, CE_{it}, ECI_{it}, GRI_{it}, FD_{it}) \tag{1}$$

The NRPI and FD series are analyzed logarithmically to estimate the parameters following the research function. In contrast, the other variables are estimated using the econometric model in Eq. (2) based on their natural structures.

$$LnNRPI_{it} = \alpha_0 + \alpha_1 FI_{it} + \alpha_2 CE_{it} + \alpha_3 ECI_{it} + \alpha_4 GRI_{it} + \alpha_5 LnFD_{it} + \mu_{it} \tag{2}$$

In Eq. (2), the constant term α_0 represents the estimation parameters $\alpha_1 - \alpha_5$, and μ_{it} represents the error term. In contrast, i and t denote the cross-sections and time in

the panel data, respectively. Innovation-based technological developments can contribute to achieving the goal of a low-carbon economy by increasing access to and use of green energy through renewable energy technologies (Fakher and Ahmed 2023). In contrast, technological innovations can have a negative impact on NRM, as they can solve some problems while creating others. Technological advances may increase the demand for new technological products, leading to higher consumption of natural resources. While technological tools may be energy-efficient, they can also lead to increased energy consumption and depletion of natural resources. Technological progress can nullify gains when it is accompanied by increased resource consumption (Midden et al. 2007). As a result, while technical advances can promote resource conservation and efficient use, they can worsen NRM due to rebound effects. We therefore hypothesize that FI, technological innovation, and CE practices may have either a negative or positive impact on NRM. On the other hand, we hypothesize that an increase in ECI has a beneficial effect on NRM, as it promotes sustainable development by enhancing skills and expertise (Wang et al. 2024a). Similarly, we assume that a well-developed financial system will enhance NRM management by playing a crucial role in financing investments, such as green technology and renewable energy.

Methodology

This study examines the NRPI function using a panel dataset of EU countries. A panel data analysis was conducted to identify the long-run relationships between the NRPI and the explanatory variables from 2004 to 2023 across. Within the methodological framework, logarithmic transformations are applied to the series, and descriptive statistics are calculated for the resulting data. To eliminate the problem of exogeneity in the estimation model, the series must be correlated. In this context, the correlation between the series was calculated considering the normal distribution. Afterwards, various tests for multicollinearity and endogeneity in the model were applied. Since EU countries are analyzed in this study, the interdependence of the countries and the heterogeneity arising from the different characteristics within the group are considered. The dependence between countries necessitated the use of second-generation unit root tests for stationarity. Following the preliminary tests for panel data analysis, cointegration tests were used to investigate whether natural resource management and the explanatory variables are cointegrated in the long run. This study analyzes twenty years of data from eight EU countries, and the panel data size cannot be defined broadly. While the Estimated Generalized Least Squares (EGLS) estimator, which provides accurate and reliable results when the panel data size is small, is used for the main results, second-generation estimators that consider cross-sectional dependence (CSD) and heterogeneity are also employed to consolidate the estimation findings.

Moreover, postestimation tests are applied to validate the estimation results. To evaluate the linkages between the NRPI and FI, CE, ECI, GRI, and FD in detail, causality relationships were investigated for eight EU countries on a panel basis. The results of the panel data analysis, conducted in accordance with the study's objectives, are presented in Fig. 2.

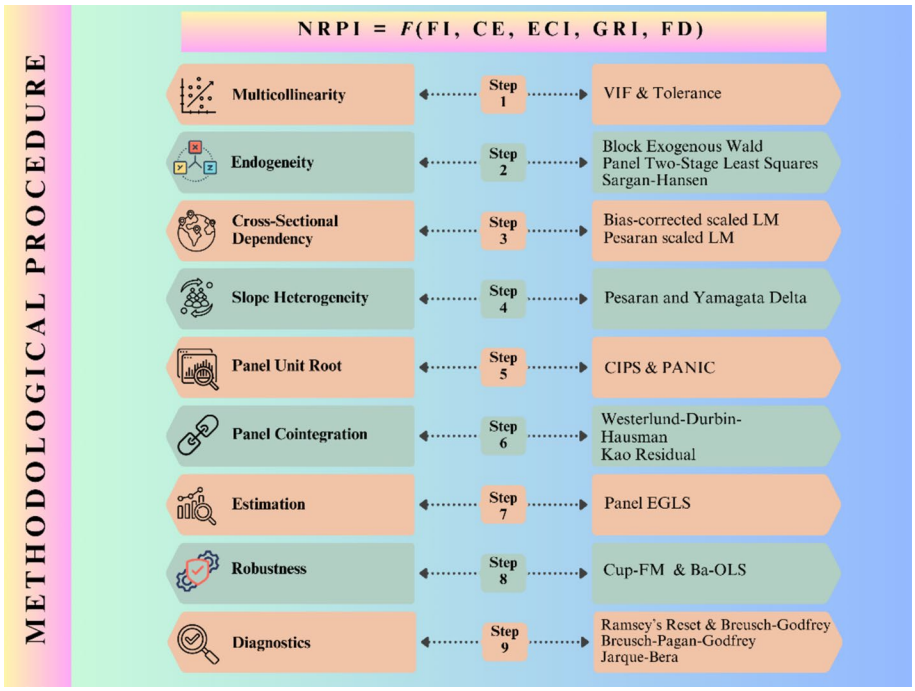


Fig. 2 Flowchart of the research method

Correlation, multicollinearity, and endogeneity evidence

Some preliminary tests should be applied to panel data analysis to accurately and reliably assess the research function. First, a normal distribution, multicollinearity, and endogeneity should be checked. Descriptive statistics were calculated for the series, and according to the kurtosis, skewness, and Jarque–Bera statistics, the normal distribution was not valid. Following this finding, the correlation between the NRPI and the explanatory variables is calculated via the Spearman test, as in Eq. (3).

$$r_s = 1 - \frac{6\sum_{i=1}^{n} d_i^2}{n\left(n^2 - 1\right)} \tag{3}$$

Equation (3) shows that x and y denote any two variables, whereas $i = 1,2,\ldots,n$, (x_i, y_i) represents the i_{th} measurement of the series. In panel regression estimation, multicollinearity and endogeneity problems may arise among the variables. Considering these problems in terms of the accuracy of the estimation results is crucial. In the present study, correlation findings for multicollinearity are supported by the variance inflation factor (VIF) and tolerance coefficients. A high level of correlation, a VIF higher than 1–5, and a tolerance value lower than 0.20 indicate multicollinearity. The VIF and tolerance coefficients are calculated via Eq. (4).

$$VIF = \frac{1}{1 - R^2} = \frac{1}{Tolerance} \tag{4}$$

The presence of statistically significant and high-level linkages between the error terms of the explanatory variables and the invalidity of the instrumental variables suggests the

presence of an endogeneity problem. The Wald test, panel two-stage least squares, and Sargan-Hansen results are considered for endogeneity in the present study. Given that there are no problems of endogeneity or multicollinearity, we can confidently state that the estimation model is accurately and reliably constructed.

Cross-sectional dependence test

There are several challenges when estimating models via longitudinal datasets, including the CSD. Failing to consider this problem before conducting empirical research may result in biased, inconsistent, and suboptimal parameters, leading to incorrect conclusions. Interdependence between EU countries is expected. CSD in a panel dataset can result from unobserved concurrent fluctuations, spatial effects, implicit economic variables, externalities, and the integration of financial and economic elements across different entities or units in the dataset. To examine the presence of CSD, this study uses the bias-corrected scaled LM and Pesaran scaled LM tests introduced by Pesaran et al. (2008) and Pesaran (2004), respectively. The null (H_0) and alternative (H_1) hypotheses of CSD are presented in Eqs. (5) and (6).

$$H_0 : \rho_{ij} = Cov\left(\mu_{it}, \mu_{jt}\right) = 0 \tag{5}$$

$$H_0 : \rho_{ij} = Cov\left(\mu_{it}, \mu_{jt}\right) \neq 0 \tag{6}$$

The panel dataset consists of closely related time and cross-sectional observations, with the time dimension exceeding the number of cross-sections. The scaled LM test statistics proposed by Pesaran (2004) for CSD are calculated via Eq. (7), whereas the bias-corrected scaled LM test statistics designed by Pesaran et al. (2008) are computed via Eq. (8).

$$LM = \sqrt{\frac{1}{N(N-1)}}\left(\sum_{i=1}^{N-1}\sum_{j=i+1}^{N}\left(T\hat{p}_{ij}^2 - 1\right)\right) \tag{7}$$

$$LM_{adj} = \sqrt{\frac{2T}{N(N-1)}}\left(\sum_{i=1}^{N-1}\sum_{j=i+1}^{N}\hat{\Psi}ij\right)\left[\frac{(T-K)\hat{\Psi}_{ij}^2 - (T-K)\hat{\Psi}_{ij}^2}{Var(T-K)\hat{\Psi}_{ij}^2}\right] \tag{8}$$

The equations representing the CSD in calculating the Lm test statistic asymptotically follow the normal distribution, with νT_{ij} representing the variance and μT_{ij} representing the mean.

Slope heterogeneity test

EU countries with geographically similar characteristics may differ in financial, economic, and social terms. The fact that these countries have different dynamics points to heterogeneity. To obtain reliable and unbiased results, it is essential to analyze the homogeneity/heterogeneity of the variables and the overall structure of the panel dataset, and to test the stationarity of the variables within the panel dataset. This helps in

selecting the appropriate tests for cointegration and causality analyses. In this study, we analyze slope heterogeneity on a variable and model basis via the delta test introduced by Swamy (1970) and later designed by Pesaran and Yamagata (2008). The delta test statistics are calculated via Eq. (9).

$$\tilde{\Delta}_{adj} = \sqrt{N}\frac{N^{-1}\hat{S} - E(Z_{it})}{\sqrt{Var}} \sim N(0,1) \tag{9}$$

In the equation, *Var(t, k)* denotes the standard error; *S* symbolizes the Swamy test statistic; *N* represents the cross-section; and *k* expresses the explanatory variable.

Panel unit root tests

To test the natural resource management function, the stationarity of the series should be checked. The estimation findings may be inconsistent and biased in panel data analyses where stationarity is not considered. In the present study, heterogeneity and CSD are proven for EU countries. When CSD is present, both parametric and nonparametric techniques are recommended for analyzing the panel data integration sequence. In panel data with evident CSD and heterogeneity, rejection of the null hypothesis cannot be considered (Choi 2001; Maddala and Wu 1999). Therefore, second-generation tests that consider heterogeneity and CSD effects should be utilized. The cross-sectional Im et al. (CIPS) test (Pesaran 2007) and Bai and Ng's (2004) PANIC are used in this study to address this issue. To determine the stationarity levels of the series, CIPS test statistics are calculated via Eqs. (10)–(12).

$$\Delta Y_{it} = \beta_i + \pi_i y_{i,t-1} + \lambda_i \bar{y}_{t-1} + \delta_i \Delta \bar{y}_t + \mu_{it} \tag{10}$$

The CADF test statistics are given by Eq. (10). Adding (*t-1*) to this equation yields Eq. (11) .

$$\Delta Y_{it} = \beta_i + \pi_i y_{i,t-1} + \lambda_i \bar{y}_{t-1} + \sum_{j=0}^{p} \delta_i \Delta \bar{y}_{t-j} + \sum_{j=1}^{p} \phi_{ij} \Delta y_{i,t-j} + \mu_{it} \tag{11}$$

The CIPS approach is performed after calculating the CADF test statistics, as shown in Eq. (2), where $\bar{y}_{t-1}$ and $\Delta y_{i,t-j}$ represent each cross-section's lagged level mean and the first difference operator, respectively.

$$CIPS = N^{-1} \sum_{i=1}^{N} \pi_i(N,T) \tag{12}$$

Equations (13) and (14) present the PANIC unit root test, which is based on the ADF procedure for panel data, to robustify the stationarity findings.

$$P_{\hat{e}}^{c} = \frac{-2\sum_{i=1}^{N} \ln P_{\hat{e}}^{c}(i) - 2N}{\sqrt{4N}} \rightarrow^{d} N(0,1) \tag{13}$$

$$P_{\hat{e}}^{T} = \frac{-2\sum_{i=1}^{N} \ln P_{\hat{e}}^{T}(i) - 2N}{\sqrt{4N}} \rightarrow^{d} N(0,1) \tag{14}$$

The equation checks whether $\rho = 0$ for all (i) or $\rho < 0$ for at least one (i).

Westerlund and Kao cointegration methods

The primary goal of this study is to investigate whether the NRPI and FI, CE, ECI, GRI, and FD variables are cointegrated in the long term. Evidence of a long-term connection between NRPI and the explanatory variables is decisive for estimating the direction and magnitude of the link. In this respect, the current study's first- and second-generation tests examine the long-term linkages between variables. First-generation panel cointegration tests employ residual-based techniques, whereas second-generation panel cointegration tests incorporate structural dynamics and account for CSD.

The present study investigates cointegration via the techniques of Westerlund (2008) and Kao (1999). Westerlund's (2008) technique assumes that the series is stationary in its first differences and, in particular, that the dependent variable has a unit root. Furthermore, this method can provide reliable results under CSD and panel heterogeneity. The Kao (1999) cointegration test also recognizes heterogeneity among cointegration vectors and is a method that addresses and accounts for panel standard errors, such as endogeneity, heteroskedasticity, and autocorrelation. In the panel data of this study, the dependent variable and other series exhibit unit roots, but become stationary after taking their first differences.

Additionally, the panel data model includes CSD and heterogeneity and accommodates possible standard errors. The panel data properties meet the conditions for applying these two tests and are appropriate. Westerlund's (2008) method is employed for the main results, in line with the properties of the panel data. To validate Westerlund's (2008) cointegration findings, a long-run cointegration test based on the residuals was employed, utilizing the Augmented Dickey-Fuller (ADF) statistic proposed by Kao (1999) for panel data. This test analyzes long-term correlations between variables. The Westerlund (2008) test employs two separate test statistics: the Durbin-Hausman panel (DHp) and Durbin-Hausman group (DH_g) statistics. The DH_g statistic yields reliable results when the autoregressive parameter is heterogeneous across groups, whereas the DH_p statistic is robust when the autoregressive parameter is homogeneous across panels. The DH_g and DH_p are calculated via Eqs. (15)–(16).

$$DH_g = \sum_{i=1}^{n} \hat{S}_i \left(\tilde{\phi}_i - \tilde{\phi}_i\right)^2 \sum_{t=2}^{T} \hat{e}_{it}^2 - 1 \tag{15}$$

$$DH_P = \hat{S}_n \left(\tilde{\phi} - \tilde{\phi}\right)^2 \sum_{i=1}^{n} \sum_{t=2}^{T} \hat{e}_{it-1}^2 \tag{16}$$

In the equations above, S_i denotes the variance, ϕ denotes the cointegration parameter, and e represents the error term. According to Kao (1999), the panel cointegration ADF test statistics are calculated using Eq. (17).

$$ADF = t_{\overline{p}} + \sqrt{\frac{6N\hat{\sigma}_r}{2\hat{\sigma}_{0r}}} / \sqrt{\frac{\sigma_{0r}^{\hat{2}}}{2\sigma_r^{\hat{2}}} + 3\sigma_r^{\hat{2}} r_r^2 / 10\left(6\hat{\sigma}_{0r}\right)} \tag{17}$$

Panel EGLS estimation

This study employs panel data analysis, which enables the simultaneous examination of time and cross-sectional data for EU states to investigate the factors influencing natural resource management. This method examines the econometric function derived from the study's theoretical framework. Once the cointegration relationship between the variables is obtained, the long-run dynamic relationship must be quantified by estimating the coefficients. After finding that the NRPI and the explanatory variables are cointegrated, the elasticity coefficients for the impacts of FI, CE, ECI, GRI, and FD on the NRPI are estimated. The estimated generalized least squares (EGLS) method is a common and preferred approach for estimating elasticity coefficients. EGLS is applied to the main results of the present study. Panel EGLS is an estimation method that considers random and fixed effects in panel data. This method accounts for the heterogeneity and autocorrelation that arise when OLS is employed. GLS is considered an unbiased β model with less variance, providing efficient and consistent estimates of autocorrelation, heteroscedasticity, and CSD (Sugandi 2022). In the present study, we estimate the coefficients via the panel EGLS model, which provides robust and reliable results under slope heterogeneity to correct for cross-sectional dependence. Equation (18) shows the general model of Panel-EGLS.

$$Y = X\varphi + \varepsilon \tag{18}$$

The given equation expresses the relationship between the explained variable, represented by a vector Y of size $n \times 1$, and the independent variables, defined by a matrix X of size $n \times p$. The unknown parameter is denoted by ϕ. To estimate the covariance matrix (Ω), we can use the unbiased estimator obtained via Eq. (19).

$$\hat{\Omega} = (N-1)^{-1} \sum_{i=1}^{N} \left(y_{it} - \overline{y}\right)\left(y_{it} - \overline{y}\right)^T \tag{19}$$

The GLS estimates for the value of ϕ are derived via Eq. (20).

$$\varphi_{BLGS} = \left(X' \Omega^{-1} X\right)^{-1} X' \Omega^{-1} Y \tag{20}$$

where X' Ω^{-1}X is nonsingular.

Eliminating standard errors that may arise in panel data estimations is crucial for the reliability and stability of the estimation findings. In the present study, the estimation results are evaluated in terms of reliability and stability through various statistical tests, including the Breusch–Godfrey serial correlation LM test, the Breusch-Pagan–Godfrey heteroscedasticity test, the Jarque–Bera normality test, and the Ramsey reset test.

Robustness tests

As an additional check to complement the results and ensure consistency and validity, we also check the robustness of the results by applying two specific techniques: Westerlund (2007), Bias-Adjusted (Ba-OLS), and Bai et al. (2006), which are continuously updated, Fully Modified (Cup-FM), to correct standard errors. These methods are robust enough to overcome panel standard errors, such as CSD and slope heterogeneity. Standard errors often provide valid statistical inferences when some assumptions in a panel regression model are violated. Since cross-sectional units are not randomly selected in macroeconometric panel datasets, the problem of spatial dependence may arise. Moreover, countries with different economic, financial, and energy demand characteristics, such as the EU countries that constitute the panel data of this study, are not homogeneous. Accordingly, the empirical model cannot be estimated correctly when the cross sections within a group are correlated and have a heterogeneous structure.

The current study's panel dataset has CSD and heterogeneity. Accordingly, the estimation results are robust to the use of second-generation Cup-FM and Ba-OLS estimators, which can provide accurate results under CSD and heterogeneity in panel data. The Cup-FM estimator, used to enhance the robustness of the estimation results, assumes that the series is stationary in the same order and provides consistent and efficient results in models with multifactor variables of different degrees of integration. Cup-FM estimation is performed via Eqs. (21)–(24).

$$E_{jt} = \gamma Y_{jt}^2 + \beta Y_{jt} + \varepsilon_{jt} \rightarrow \underset{jt}{x'} + \varepsilon_{jt} \tag{21}$$

where

$$\varepsilon_{jt} = \lambda_j' F_t + u_{jt} \tag{22}$$

$$x_{jt}' = x_{jt-1}' + \varepsilon_{jt} \tag{23}$$

$$F_t = F_{t-1} + \eta_{jt} \tag{24}$$

Cup-FM corrects for standard errors in panel data via two-equation iterations and assumes that the number of common factors represented by k is known; however, this approach needs improvement. The Ba-OLS technique is based on a bias correction methodology that determines the number of common factors. This technique estimates k via the information criterion, as expressed in Eq. (25).

$$\hat{k} = \min\left(IC(k)\right), 1 < k < k_{max} \tag{25}$$

In the above equation, *IC(k)* represents the information criterion, whereas the number of common factors is determined via the Bayes information criterion as in Eq. (26).

$$BIC(k) = V(k) + kV(k_{max)}\left(\frac{(N+T-k)ln(NT)}{NT}\right) \tag{26}$$

In Eq. (26), k is the estimated variance based on the factor k, while the minimized information criterion determines the optimal value of the parameter.

Emirmahmutoglu–Kose panel causality test

Following the estimation of the elasticity coefficients of the explanatory variables on the NRPI, unidirectional or bidirectional causal connections between the variables are examined using Emirmahmutoglu and Kose (2011) (E–K) to elaborate on the estimation findings and determine the link between the explanatory variables and the NRPI. The E–K addresses the complexities inherent in panel data models with varying characteristics. This method is a new-generation panel causality approach that extends causality tests to panel data, minimizes biases in unit roots and cointegration, and can provide robust results under CSD in heterogeneous panel datasets. Since the panel data of the study exhibit CSD and heterogeneity, the E-K method is an appropriate approach for determining causality results.

This test evaluates CSD and heterogeneity by calculating the Fisher test statistic depending on the LA-VAR approach. Equation (27) estimates the null hypothesis H_0: $A_{1,2,ij}=0$ with (k_i+dmax), whereas Eq. (28) calculates the Fisher test statistics.

$$z_{i,t} = \mu_i + A_{i1}z_{i,t-1} + ... + A_{ik_i}z_{i,t-k_{i'}} + \sum_{1=k_i+1}^{k_i+d\max_i} A_{i1}z_{i,t-1} + u_{i,t} \tag{27}$$

$$\hat{\lambda} = -2\sum_{i=1}^{N} \ln(p_i) i = 1, 2, \ldots, N \tag{28}$$

Equations (29) and (30) compute the Fisher test statistics calculated via the bootstrap procedure.

$$x_{i,t} = \mu_i^x + \sum_{j=1}^{k_i+dmax_i} A_{11,ij}x_{i,t-j} + \sum_{j=1}^{k_i+dmax_i} A_{12,ij}y_{i,t-j} + u_{i,t}^x \tag{29}$$

$$y_{i,t} = \mu_i^y + \sum_{j=1}^{k_i+dmax_i} A_{21,ij}x_{i,t-j} + \sum_{j=1}^{k_i+dmax_i} A_{22,ij}y_{i,t-j} + u_{i,t}^y \tag{30}$$

In the VAR model with k_i+dmax lags, $dmax_i$ represents the maximum degree of integration for cross-sections.

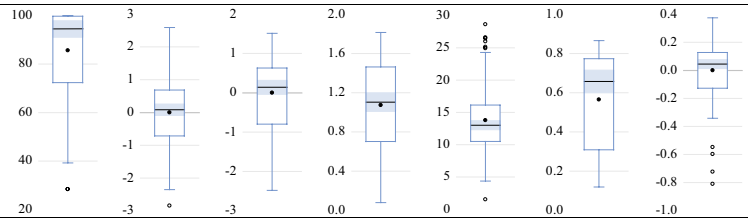
Results and discussion

Preliminary tests

Statistical significance levels, mean values, standard deviations, skewness, kurtosis coefficients, and JB test statistics were calculated for each parameter in EU countries (Table 2). The mean values for NRPI, ECI, GRI, and FD are 85.562, 1.074, 13.804, and 0.656, respectively. The deviation for NRPI is 16.762, whereas the lowest deviation recorded for FD is 0.229. After evaluating the skewness coefficients, the GRI series exhibited a left-skewed distribution, whereas the other series displayed right-skewed distributions. Additionally, kurtosis coefficients indicate that all series have a skewed distribution. Based on the JB test statistics, all series except FI reject the assumption of

Table 2 Descriptive statistics and plots

Box plots



Stats.	NRPI	FI	CE	ECI	GRI	FD	ET
Mean	85.652	0.000	0.000	1.074	13.804	0.565	-4E-16
Median	94.598	0.090	0.145	1.107	13.056	0.659	0.046
Maximum	100.000	2.592	1.510	1.815	28.645	0.866	0.374
Minimum	28.413	-2.841	-2.479	0.080	1.533	0.119	-0.810
Std. Dev.	16.762	1.006	0.842	0.453	4.943	0.229	0.196
Skewness	-1.153	-0.089	-0.436	-0.347	0.641	-0.431	-1.015
Kurtosis	3.672	3.038	2.346	2.057	3.367	1.549	5.067
Jarque-Bera	38.434***	0.219	7.925**	9.134**	11.870***	18.993***	55.951***
Observations	160	160	160	160	160	160	160

Scatters plots

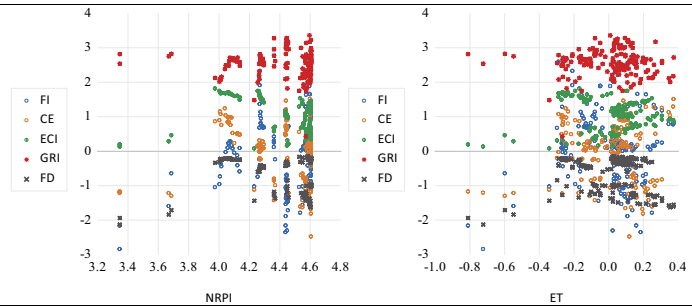


Table 3 Heat-plot of Spearman correlation matrix

Mean VIF (1.778)			ET	NRPI	FI	CE	ECI	GRI	FD
1/VIF	VIF	ET	1.000	0.577	-0.170	-0.065	-0.150	-0.081	-0.090
0.856	1.169	NRPI	0.577	1.000	-0.242	-0.213	0.445	-0.056	0.055
0.930	1.076	FI	-0.170	-0.242	1.000	-0.020	-0.154	0.059	-0.414
0.374	2.673	CE	-0.065	-0.213	-0.020	1.000	0.190	0.065	0.141
0.928	1.078	ECI	-0.150	0.445	-0.154	0.190	1.000	-0.254	0.692
0.346	2.892	GRI	-0.081	-0.056	0.059	0.065	-0.254	1.000	-0.113
		FD	-0.090	0.055	-0.414	0.141	0.692	-0.113	1.000
			ET	NRPI	FI	CE	ECI	GRI	FD

a normal distribution. Box charts for each series's quarterly mean, dispersion, and variability indicators are presented in Table 2, and scatter plots based on the model are generated. As a result, while the presence of asymmetric impacts in the series is detected, it is observed that they do not have a normal distribution. Descriptive statistics provide the opportunity for additional analyses.

Multicollinearity–endogeneity evidence

The present study conducts VIF and tolerance tests for multicollinearity. Moreover, panel two-stage least squares, block exogenous Wald, and Sargan-Hansen tests for endogeneity are applied. Descriptive statistical calculations revealed that the series was not normally distributed. Hence, Spearman's correlation test is used when there is no normal distribution and provides results that are not affected by deviation. Bivariate heat plots of correlation matrices for logarithmic and proportional variables

Table 4 Endogeneity test results

Variable	Coefficient	*t*-Statistic	Prob
Panel two-stage least squares			
FI^{RESID}	−0.011	−0.864	0.388
CE^{RESID}	−0.005	−0.361	0.718
ECI^{RESID}	0.014	0.737	0.462
GRI^{RESID}	−0.010	−0.714	0.475
FD^{RESID}	0.054	0.775	0.439
Hypothesis—H_0: exogenous		**$X^2(1)$**	**Prob**
Block exogenous Wald test			
FI	CE	2.169	0.141
	ECI	1.497	0.221
	GRI	0.737	0.390
	FD	1.345	0.246
CE	FI	0.594	0.440
	ECI	1.007	0.315
	GRI	0.418	0.517
	FD	0.221	0.637
ECI	CE	0.026	0.871
	FI	1.497	0.221
	GRI	0.565	0.452
	FD	0.949	0.330
GRI	CE	0.224	0.636
	ECI	0.211	0.646
	FI	0.442	0.506
	FD	0.453	0.501
FD	CE	1.703	0.192
	ECI	0.310	0.577
	GRI	0.301	0.583
	FI	1.272	0.259
Sargan-Hansen Test for Exogeneity of Instruments			
Instrument specification:	Instrument validity	Sargan–Hansen J statistic	Prob(J-statistic)
@DYN(NRPI,-2) FI(-1) CE(-1) ECI(-1) GRI(-1) FD(-1)	Model	2.640	0.267

H_0: The instruments used in this model are valid

are shown in Table 3. The highest correlation coefficient between the explanatory variables is 0.692, which is observed between ECI and FD. Another correlation test result is between the explanatory variables and the error term. The highest estimated correlation coefficient between the error term and FI is 0.170. Therefore, we employed the VIF test to assess the robustness of the correlation findings, yielding an average VIF of 1.778. These findings substantiate the observations in the scatter plots. Estimation findings via the panel two-stage least squares technique indicate no endogeneity issues in the explanatory variables. Block exogenous Wald and Sargan-Hansen tests confirm the estimation findings and demonstrate that FI, CE, ECI, GRI, and FD are valid non-endogenous instrumental variables (Table 4). The results indicate no issues of multicollinearity or endogeneity in the econometric model.

Table 5 Cross-sectional dependence and slope heterogeneity test results

Variable	Pesaran scaled LM		Bias-corrected scaled LM		Delta tests			
	Stat	Prob	Stat	Prob	$\tilde{\Delta}$	Prob	$\tilde{\Delta}_{adj}$	Prob
NRPI	8.862	0.000	4.189	0.000	9.268	0.000	10.053	0.000
FI	3.025	0.001	6.678	0.000	−0.977	0.836	−1.060	0.855
CE	2.338	0.010	4.814	0.000	0.845	0.199	0.917	0.180
ECI	20.367	0.000	3.600	0.000	−0.486	0.687	−0.528	0.701
GRI	27.591	0.000	27.380	0.000	1.817	0.035	1.971	0.024
FD	5.857	0.000	20.014	0.000	1.598	0.055	1.733	0.042
Model	2.087	0.018	10.287	0.000	8.704	0.000	10.660	0.000

Panel cross-sectional dependence and slope heterogeneity results

In this study, we employed two distinct CSD analyses proposed by Pesaran (2004) and Pesaran et al. (2008) to investigate the potential interrelationships among countries with high levels of NRM. The variables under examination include the NRPI, CE, FI, ECI, GRI, and FD. Additionally, a CSD analysis was conducted on the model. The findings of the two CSD techniques are presented in Table 5. The findings from these techniques are statistically significant, providing clear evidence of the presence of CSD. Consequently, we decisively reject the null hypothesis, which posits no evidence of CSD in the dataset. This observation highlights the presence of CSD among the variables studied, indicating that disturbances in NRM in one country can affect other countries.

When analyzing the dataset, it is essential to check for slope heterogeneity, as the slope coefficients may differ across sectional units. Although countries with NRM are interdependent, they can still differ significantly in terms of economic size, macroeconomic factors, and renewable energy structures, among other aspects. Therefore, addressing concerns regarding slope heterogeneity is essential. This study examines slope heterogeneity using the delta test, as designed by Pesaran and Yamagata (2008), which is also presented in Table 5. The null hypothesis that the slopes are homogenized is rejected, as the findings show that the NRPI, GRI, and FD exhibit statistically significant evidence of slope heterogeneity. However, the null hypothesis for FI, CE, and ECI could not be rejected. Additionally, the delta test statistics calculated for the entire dataset provide evidence of heterogeneity in the model.

Panel unit root evidence

After addressing issues with slope heterogeneity and CSD in the dataset, checking for stationarity in all variables is essential. This step is crucial for ensuring the reliability and suitability of the regression estimation methodology, mainly when second-generation panel unit root approaches are used. In the presence of heterogeneity and CSD in both the variables and the panel data, Pesaran's (2007) CIPS and PANIC unit root tests are especially beneficial for obtaining reliable coefficients of determination. Table 6 presents the stationarity test results, which assess the stationarity of the variables. Notably, the null hypothesis of no CSD is definitively rejected for all variables for EU countries with

Table 6 Results of unit root test

Variables	CIPS		PANIC			
	t-stat	Prob	PCe Choi	Prob	PCe MW	Prob
Intercept						
NRPI	−1.591	≥0.10	−0.523	0.699	13.040	0.669
ΔNRPI	−3.209	<0.01	4.866	0.000	43.529	0.000
FI	−2.043	≥0.10	0.484	0.314	18.743	0.282
ΔFI	−2.768	<0.01	4.897	0.000	43.701	0.000
CE	−1.895	≥0.10	−0.777	0.781	11.603	0.770
ΔCE	−3.286	<0.01	4.402	0.000	40.902	0.000
ECI	−2.192	≥0.10	−0.366	0.643	13.928	0.604
ΔECI	−2.749	<0.01	4.183	0.000	39.667	0.000
GRI	−1.883	≥0.10	1.161	0.122	22.569	0.125
ΔGRI	−3.167	<0.01	4.030	0.000	38.800	0.001
FD	−1.764	≥0.10	−0.986	0.838	10.419	0.843
ΔFD	−3.650	<0.01	4.121	0.000	39.313	0.001
Critical values		1%(-2.61) 5%(-2.35) 10%(-2.21)				
Variables	**CIPS**		**PANIC**			
	***t*-stat**	**Prob**	**PCe Choi**	**Prob**	**PCe MW**	**Prob**
Intercept and trend						
NRPI	−1.478	≥0.10	−0.171	0.568	15.031	0.522
ΔNRPI	−3.420	<0.01	3.931	0.000	38.240	0.001
FI	−2.035	≥0.10	−0.222	0.588	14.741	0.543
ΔFI	−3.631	<0.01	3.234	0.000	34.294	0.005
CE	−1.613	≥0.10	0.512	0.304	18.901	0.273
ΔCE	−3.424	<0.01	3.184	0.000	34.010	0.005
ECI	−2.533	≥0.10	−0.291	0.614	14.349	0.572
ΔECI	−3.521	<0.01	4.906	0.000	43.757	0.000
GRI	−2.314	≥0.10	0.509	0.305	18.880	0.275
ΔGRI	−3.827	<0.01	7.053	0.000	55.897	0.000
FD	−2.076	≥0.10	−1.629	0.948	6.784	0.977
ΔFD	−3.887	<0.01	3.196	0.000	34.079	0.005
Critical values		1%(-3.17) 5%(-2.89) 10%(-2.74)				

NRM. The CIPS and PANIC unit root test results reveal that the hypothesis that a time series remains unchanged at I(0) is not supported for any variable. Therefore, it is clear that all variables are stationary at I(1) after first differencing.

Panel cointegration evidence by Westerlund and Kao

After assessing the degree of time series integration, this study utilizes the panel cointegration test introduced by Westerlund (2008). This technique can address the CSD and slope heterogeneity of data. Additionally, the residual-based long-run cointegration test established by Kao (1999) complements the findings from the Westerlund test. The results of these cointegration approaches are presented in Table 7. With respect to the NRM function, DH_g is significant at the 1% level, and DHp is significant at the 5% level. Similarly, the Kao test results corroborate the findings of

Table 7 Cointegration test results

Westerlund-Durbin-Hausman			Kao Residual		
	Value	prob		*t*-stat	Prob
DH_g	−2.089	0.018	ADF	−5.682	0.000
DH_p	−1.860	0.031	Residual var	0.0032	
			HAC var	0.0023	

Table 8 Panel EGLS estimate results

Regressor	Main results		$NRPI_{it} = f(CE_{it}, FI_{it}, ECI_{it}, GRI_{it}, FD_{it})(CE_{it}, FI_{it}, ECI_{it}, GRI_{it}, FD_{it})$		
	EGLS		Post estimation/diagnostics		
	Coeff	*t*-stat	Test	F/t-stat	Prob
FI	−0.091***	−17.934	Jarque Bera	4.091	0.129
CE	−0.033***	−7.395	Ramsey's Reset	0.343	0.732
ECI	0.379***	25.535	LM	0.583	0.559
GRI	−0.070***	−8.874	BPG	0.639	0.637
FD	0.300***	12.869	adj.R^2	0.903***	

*** $p < 0.01$
** $p < 0.05$
* $p < 0.1$

the Westerlund test at the 1% significance level. This finding demonstrates that all identified explanatory variables support the long-term cointegration process in NRM, exhibiting stable movements and a high degree of confidence and linkages throughout the sample period.

Panel EGLS estimate results

The long-term elasticity coefficients of the explanatory variables with respect to NRM are estimated by applying panel EGLS. The estimation findings are robust to the CupFM and BaOLS techniques (Table 8). Diagnostic tests for autocorrelation, normal distribution, and heteroskedasticity confirmed the reliability, robustness, and consistency of the estimation results. Furthermore, the Reset test confirms that the explanatory variables against NRPI have no missing nonlinear or higher-order effects. The findings indicate that the model's explanatory power is 0.903. Accordingly, a 1% increase in FI decreases NRPI by 0.091. In other words, FI does not improve NRM in EU countries. This can be explained by the fact that FI devices and infrastructure are energy-intensive. FI can increase e-waste and energy consumption (Lu et al. 2023).

In contrast, the literature also suggests that FI can enhance energy efficiency by promoting green finance, and that FI's ability to optimize resource use can improve NRM (Tan et al. 2023; Zhang et al. 2024b). Our findings suggest that this effect does not apply to EU countries during the relevant period. Therefore, policymakers need to assess these effects. However, the FI, the European Green Deal (2020), and the 8th Environmental Action Program (2022) do not consider the potential negative impact of financial innovations on the environment. According to the European Green Deal (2020), supporting the development of supercomputers, clouds, ultrafast networks, and artificial intelligence

solutions is expected to contribute to conserving natural resources by facilitating digital transformation. However, it does not consider that the increased use of financial technologies resulting from financial innovations could lead to increased energy consumption and, consequently, environmental degradation. Therefore, it is recommended that policymakers in the EU consider that financial innovations, particularly financial technologies, can increase energy consumption and lead to environmental degradation. In this context, policymakers should provide incentives for financial innovations that increase energy efficiency while imposing additional taxes on financial technologies that lead to excessive energy consumption. Therefore, it is crucial to consider the impacts of financial innovations in the 8th Environmental Action Program (2022). Hence, ensuring that the technologies used for financial development are energy efficient and promoting green finance initiatives is essential.

Similarly, a 1% increase in the GRI reduces the NRPI by 0.070. This result can be explained by technological progress contributing to environmental degradation through the rebound effect. Chu et al. (2022) argued that the discovery and use of cleaner and more accessible energy sources stimulate both production and consumption by Lowering energy prices, leading to the faster depletion of natural resources. Our findings show a rebound effect of technological progress in EU countries. However, in the 8th Environmental Action Program (2022), technological innovations are assumed to contribute to natural resource protection and environmental sustainability through resource efficiency, while the rebound effect is overlooked. However, to prevent green technologies from increasing demand by reducing energy costs and thus accelerating the consumption of natural resources, policymakers are encouraged to develop policies to increase resource efficiency and reduce the consumption of natural resources. In this context, additional taxes should be imposed on resource consumption above a certain threshold. Therefore, these countries can ensure better resource management by creating policy instruments that take into account the rebound effect.

A 1% increase in CE results in a 0.033 decrease in NRPI. CE policies are intensively implemented in the EU. The European Commission adopted the CE model through the CE Action Plan (CEAP-1) in 2015 and adopted the European Green Deal in 2020, of which the CE is a key component. However, our findings show that CE practices do not improve resource management. This suggests a rebound effect related to unexpected outcomes and the European Green Deal's inability to achieve the desired results. This finding that the circular economy fails to ensure the conservation of natural resources suggests that the policy emphasized in the European Green Deal, which aims to increase consumer demand for recycled products, is ineffective. Accordingly, consumers' demand for recycled products creates additional consumption, leading to further depletion of natural resources through the rebound effect. This evidence suggests that the policy outlined in the European Green Deal requires revision. The results also show that the textiles, construction, electronics, and plastics sectors, where the circular economy is estimated to effectively achieve the carbon neutrality targets in the European Green Deal, do not provide the expected benefits. This result highlights the need to investigate different sectors where the impact of the circular economy can be transformed into a positive economy. Therefore, it is essential to revise the target of doubling the circular economy mentioned in the 8th Environmental Action Program (2022) and monitored in

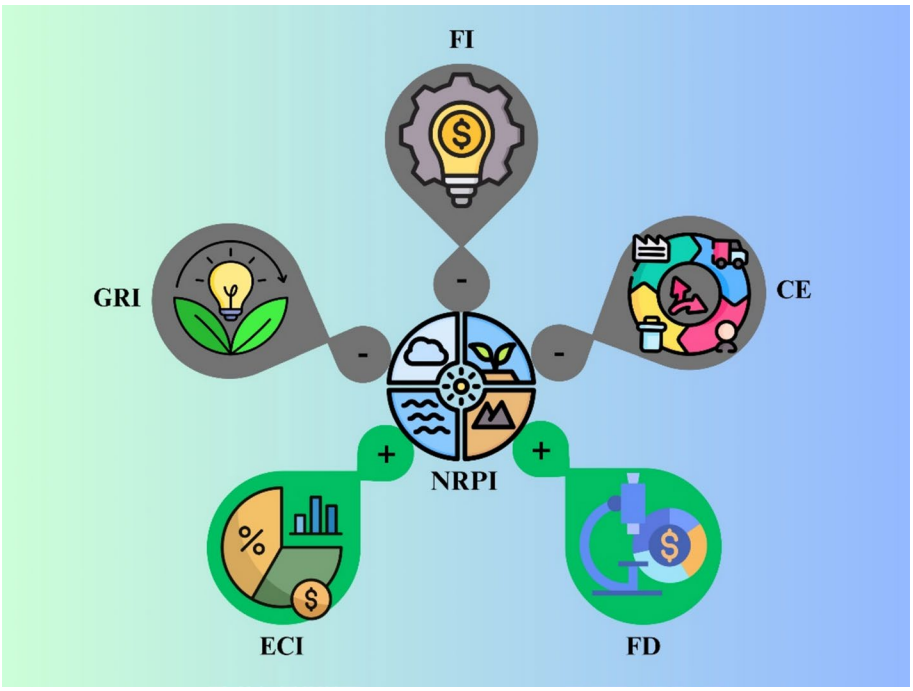


Fig. 3 Graphical Summary of results for EU countries

the Environmental Action Program Monitoring Report (2023) by reconsidering waste in the context of the circular economy and considering the rebound effect.

On the other hand, a 1% increase in ECI and FD increases NRPI by 0.379 and 0.300, respectively. Increased entrepreneurial activity is often associated with technological innovation, and new entrepreneurs are under pressure to produce more functional and cost-efficient products. This incentivizes entrepreneurs to develop new products that are less reliant on natural resources and avoid expensive production methods. Hence, increased EC will lead to higher-quality production systems (Canh et al. 2020). EC can contribute to NRM in these countries by supporting Research and Development (R&D) activities.

Similarly, FD contributes positively to natural resource conservation in these countries. A developed financial sector can incentivize financing for the renewable energy sector at lower costs, leading to increased investment and energy demand. Developed capital markets play a critical role in mitigating liquidity risk for firms and facilitating long-term investments in energy-efficient technologies.

Additionally, FD supports redirecting funds from traditional, less efficient energy sources toward renewable energy projects. Given the significant capital requirements of the renewable energy sector, robust financial markets are essential. They provide easier access to equity financing and debt, promoting the growth and development of renewable energy initiatives (Kim and Park 2016; Anton and Nucu 2020). Therefore, FD in EU countries can contribute to the conservation of natural resources. The effects of the explanatory variables on NRPI are shown in Fig. 3.

Table 9 Robustness test results

	CupFM		BaOLS	
Regressor	**coeff**	***t*-stat**	**Coeff**	***t*-stat**
FI	−0.077***	−4.566	−0.079***	−4.778
CE	−0.043***	−3.794	−0.047***	−3.946
ECI	0.486***	8.423	0.428***	8.058
GRI	−0.083***	−5.007	−0.089***	−5.038
FD	0.043***	4.019	0.040***	3.940

*** $p < 0.01$

** $p < 0.05$

* $p < 0.1$

Robustness check

We use the Cup-FM and Ba-OLS estimators to measure the key explanatory variables to test the robustness of our proposed long-term empirical results. The robustness results of the baseline model from the Cup-FM and Ba-OLS tests are presented in Table 9. The Cup-FM and Ba-OLS estimates confirm positive and significant relationships between NRPI and ECI and FD, as well as negative and significant relationships between NRPI and FI, CE, and GRI at the 1% significance level for EU countries. Although the estimated coefficients of Cup-FM and Ba-OLS are numerically slightly different from the EGLS coefficients, overall, the Cup-FM and Ba-OLS results confirm that there is still a significant relationship between NRPI and its determinants in the main specifications, and that the sign of the parameters remains unchanged. These findings are consistent with our baseline model results in Table 8. Our main empirical results remain robust and meaningful even under new model specifications and alternative model choices.

Furthermore, after accounting for CSD with heterogeneous effects across panel members in the modeling, the robustness check results support the baseline model estimates. Thus, the Cup-FM and Ba-OLS robustness results overcome the potential biases of CSD and heterogeneity and are effective as a robustness testing procedure (Guzel et al. 2021; Yildirim et al. 2022; Olasehinde-Williams and Akadir, 2024).

Emirmahmutoglu and Kose causality evidence

This study employs the panel causality test developed by E-K and provides reliable results, even when addressing the CSD and slope heterogeneity of the data. In addition to estimating the long-term elasticity coefficient, this study examines the causal relationships. When the results for EU countries are analyzed on a group basis, it is found that there are bidirectional causal relationships between CE, ECI, FI, GRI, FD, and NRPI (Table 10). Accordingly, in this group of countries, ECI, FI, CE, and GRI may affect NRM through technological development and knowledge accumulation. However, good natural resource management requires efficient production processes and technological innovation. These processes adopted to improve NRM can also support the ECI, FI, CE, and GRI processes. Therefore, combining the NRM, ECI, FI, CE, and GRI factors is essential for achieving sustainability.

Table 10 Results of panel bootstrap causality test

Causality			Fisher stat	Boots. Prob	Finding
FI	→	NRPI	54.012	0.000	Two-way
NRPI	→	FI	57.630	0.000	
CE	→	NRPI	92.820	0.000	Two-way
NRPI	→	CE	62.049	0.000	
ECI	→	NRPI	67.507	0.000	Two-way
NRPI	→	ECI	64.639	0.000	
GRI	→	NRPI	53.543	0.000	Two-way
NRPI	→	GRI	53.329	0.000	
FD	→	NRPI	69.572	0.000	Two-way
NRPI	→	FD	66.321	0.000	
ECI	→	FI	86.059	0.000	Two-way
FI	→	ECI	98.682	0.000	
CE	→	ECI	43.851	0.000	Two-way
ECI	→	CE	40.217	0.001	
GRI	→	CE	64.889	0.000	Two-way
CE	→	GRI	60.744	0.000	
ECI	→	GRI	47.407	0.000	Two-way
GRI	→	ECI	66.385	0.000	
FI	→	CE	44.928	0.000	One-way
CE	→	FI	38.790	0.522	
FI	→	GRI	78.104	0.000	One-way
GRI	→	FI	35.751	0.706	
FI	→	FD	60.567	0.000	One-way
FD	→	FI	34.426	0.856	
FD	→	CE	44.715	0.000	One-way
CE	→	FD	20.164	0.976	
FD	→	ECI	51.671	0.000	One-way
ECI	→	FD	20.732	0.930	
FD	→	GRI	55.550	0.000	One-way
GRI	→	FD	24.456	0.926	

Similarly, a sound financial system can provide resources for investments that make the management of natural resources efficient. Conversely, successful NRM can encourage the development of financial instruments in this area, channelling more resources. Moreover, the results show bidirectional causality between the ECI and FI, the ECI and CE, the GRI and ECI, and the GRI and CE. Complex economies with high knowledge, skills, and technological diversity create the necessary infrastructure and competencies for innovation, as they accumulate more knowledge and skills. On the other hand, investments in innovation can increase economic complexity by supporting the production of more sophisticated goods and services. Thus, there is a feedback loop between economic complexity and innovation. It is essential to consider these relationships for successful NRM.

Moreover, unidirectional causal relationships have been discovered from FI to CE, FI to GRI, FI to FD, FD to CE, FD to ECI, and FD to GRI. FIs can expand the financial system by encouraging greater participation in financial markets through new products and services. Additionally, developing financial systems can provide resources and funding

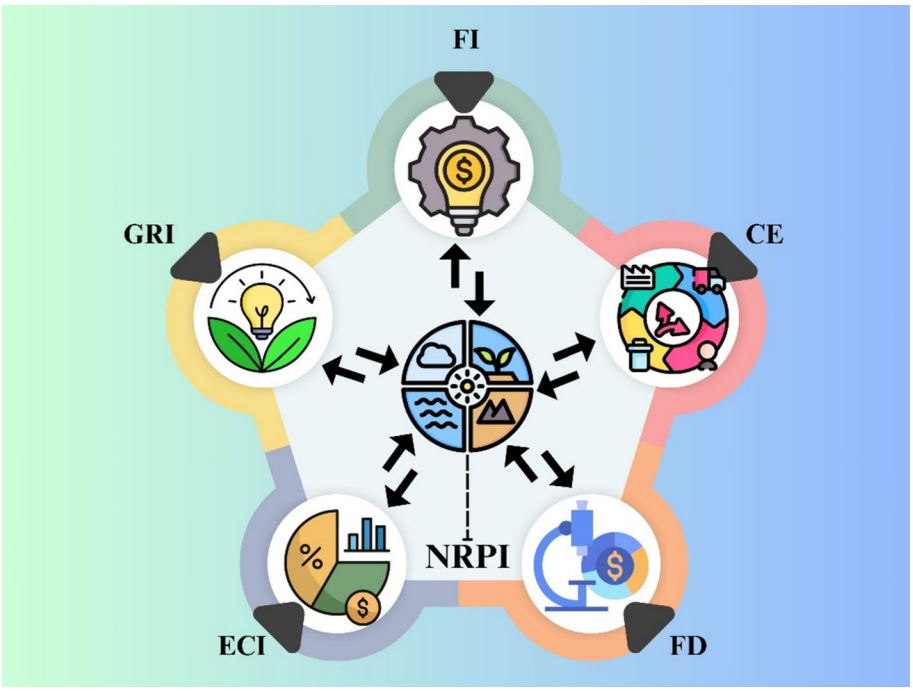


Fig. 4 Panel Fisher causality relation flow

for the development of environmentally friendly technologies. This group of countries needs interactive policies to achieve NRM and SDGs. The causal relationships between the explanatory variables and NRM for EU countries are shown in Fig. 4.

Conclusion and discussion

Economic growth, which is dependent on limited resources, leads to the rapid depletion of natural resources and increased environmental degradation. Sustainable economic growth is one of the main objectives of the SDGs. In this context, the aim is to decouple economic growth from environmental degradation. In line with this goal, there is a need to develop policies for conserving and effectively managing natural resources. In line with these policies, achieving economic growth without causing environmental degradation is possible. Although the EU has developed multifaceted policies to protect natural resources, it is rapidly depleting them due to its economy, which is based on the linear growth model. EU member states consume much more than their share of the global biocapacity. Therefore, it is crucial to identify the factors that affect NRM policies. This study analyzes the effects of EC, CE, FI, and GRI on NRM for EU member states. To the best of our knowledge, this is the first study to examine the impacts of EC, CE, FI, and GRI on NRM. The findings and discussions from the study are presented under subheadings.

Summary

This study aims to analyze the effects of EC, CE, FI, and GRI on NRM, contributing to the development of effective and efficient policies in EU countries where a

significant portion of natural resources is consumed. For this purpose, panel growth and panel causality analyses were conducted in EU countries where data are available for the longest period. Panel causality analyses were conducted on an EU basis to reveal short-run relationships. The study reached the following conclusions. 1) CE, GRI, and FI worsen NRM through the rebound effect. 2) EC and FD improve NRM. 3) There is a bidirectional causal relationship between CE, ECI, FI, GRI, FD, and NRPI. Therefore, a feedback loop exists between these variables. 4) There are bidirectional causal connections between ECI and FI, ECI and CE, GRI and ECI, GRI and CE, and unidirectional causal relationships from FI to CE, FI to GRI, FI to FD, FD to CE, FD to ECI, and FD to GRI.

Impact of economic complexity on natural resource management

The findings of this study suggest that an increase in EC is associated with an improvement in NRM. EU countries focus on increasing their product competitiveness to achieve sustainable economic growth. This creates a trend toward the creation of new and advanced knowledge-intensive products. However, with increasing global competition, the complexity of products is also on the rise. As exported products become more complex and sophisticated, they carry the risk of increasing energy demand, thereby increasing energy intensity and pollution (Neagu 2019). However, ECs can support R&D activities and new technologies, providing incentives to produce new products that are less dependent on natural resources. Our findings are consistent with this view. Therefore, increasing EC in EU countries is likely to lead to the development of efficient NRM techniques. Additionally, the Fisher causality analysis findings suggest a bidirectional relationship between EC and NRM. In other words, the EC enhances the effectiveness of NRM, while these policies and strategies encourage businesses in the EU to produce sophisticated products.

Moreover, the causality findings revealed bidirectional causal connections between EC and FI, between EC and CE, and between GRI and EC. The EC has a strong relationship with trade in green products and technologies, and countries with high levels of complexity tend to have an advantage in producing green products. EC encourages innovation by increasing a country's level of expertise (Mealy and Teytelboym 2022; Wang et al. 2024b). The findings support this view. However, the CE, FI, and development of green technologies can increase EC by enabling the creation of new products and processes. Therefore, the relationships of CE, FI, and GRI with EC should be considered when developing policies on the supporting role of EC in improving NRM in EU countries.

Impact of financial development on natural resource management

The findings show that FD improves NRM. A developed financial sector can support renewable energy by providing the necessary funds to promote energy-efficient technologies, contributing to the conservation and sustainability of natural resources. However, the causality findings support this view and reveal unidirectional causal relationships from FD to CE, GRI, and EC. A unidirectional causal connection is also found between FI and FD. FD is recognized as a critical element for modern economies, as the diversity of services offered by financial intermediaries can stimulate entrepreneurial and innovative activities. When FD and innovation come together, they can directly contribute by

increasing the complexity of the economic system through the development of new and alternative products (Nguyen et al. 2020). In this context, the role of innovation should also be considered in improving the impact of FD on NRM in EU countries.

Impact of innovation on natural resource management

Another study finding concerns the impact of CE practices on NRM policies. The EU has developed numerous policies and strategies to transition to a CE model. First, the EU launched the CE Action Plan (CEAP-1) in 2015, initiating the process of CE transformation. In 2020, it adopted the European Green Deal, the core component of the CE, and adopted a new CE action plan (CEAP-2). It has also introduced several restrictive regulations on packaging waste, plastic use, waste batteries, and hazardous materials. Despite numerous legislative measures, the EU's transition to a CE has not succeeded. In other words, the EU has set an example for other countries in developing CE policies, but has made little effort to move beyond the linear economy model, which is based on the extraction and use of natural resources (Friant et al. 2021). The results suggest that CE practices reduce the effectiveness of NRM. This shows the rebound effect, which refers to unexpected results. However, more efficient energy management is achieved through CE practices; consumption and production increase when the system does not adapt to this change.

Similarly, the findings show that GRI does not improve NRM in EU countries. The rebound effect of technological progress can explain this result. The rebound effect occurs when consumers use savings from efficiency investments to increase their overall consumption. This suggests that market forces and humanity's seemingly boundless capacity for consumption will leverage new technology to transform more resources into an excellent array of activities, functions, services, and products (Herring and Roy 2007). This results in greater energy demand and consumption of natural resources. Finally, FI has been found to have a negative effect on NRM. Lu et al. (2023) stated that FI could lead to e-waste and increased energy consumption. In contrast, Khan et al. (2024) suggested that innovations, which assume that FI will reduce costs in natural resource-based energy firms, may result in higher energy demand and environmental pollution at low energy prices for consumers. Our findings support these findings.

However, according to the causality analysis, there are unidirectional causal relationships from FI to CE, unidirectional causal relationships from FI to GRI, and bidirectional causal relationships between GRI and CE. FI can reduce the information asymmetry faced by the GRI and contribute to the efficient development of green innovation (Yuan et al. 2021; Wang and Li 2024), as well as the financing of CE projects. However, while technological advances can increase energy efficiency at the micro level, they can also lead to increased energy consumption at the macro level (Chu 2022). In other words, when policies are developed and implemented in EU countries, considering the rebound effect of technological advances, their impact on NRM can be positive.

Policy implications

Based on these findings and discussions, this study has several policy implications for EU countries. Firstly, the findings confirm that the EC has developed NRM in EU countries. Therefore, these countries have taken into account the knowledge-based complexity of

natural resource conservation. Structured policies can be developed to mitigate the positive effects of EC on NRM. Policymakers should create incentives that enable enterprises that generate high export revenues by producing sophisticated products to channel a certain percentage of their revenues into renewable energy investments. In this way, the greater the exports of sophisticated products, the more natural resources, such as fossil fuels, can be conserved. Moreover, the practical impact of EC on NRM can be further enhanced by utilizing existing capacities, creating new products, and reducing energy consumption. The depletion of natural resources can be avoided by introducing tax exemptions for energy-efficient investments by enterprises producing sophisticated products. In addition, policymakers could develop tightened regulations to channel the increased prosperity of exporting sophisticated products into producing environmentally friendly products that consume fewer resources. Again, providing incentives and policies to increase EC in these countries can reduce natural resource dependence. However, given the bidirectional causal relationships between ECs and NRM, it is essential to implement policies that interact between ECs and NRM in these countries to achieve sustainability goals. For NRM to be effective, supportive processes such as R&D investment and sustainability of EC are crucial. Secondly, in EU countries, the CE has been found to have a rebound effect on NRM. Despite numerous legal regulations, the expected success of CE implementation may not be achieved without adaptation to local contexts and intergovernmental cooperation on standard measures in these countries. Therefore, EU countries should not only focus on energy savings and job creation in CE development, but also implement more comprehensive policies for sustainable resource management and environmental protection to avoid the rebound effect of the CE. To mitigate the rebound effect of the CE, these countries should monitor the benefits and costs of the CE, promote renewable energy sources, offer sector-based incentives for new technologies that reduce demand for new raw materials, and implement effective waste management strategies. By establishing legal standards and regulations for circular economy principles in this group of countries, differences in implementation between governments can be eliminated, and transparency can be ensured. In addition, resource efficiency and sustainable development can be supported by implementing integrated policies on natural resource management and CE practices.

Thirdly, Green technology has not yet improved NRM in EU countries. To counter the rebound effect of technology, it is crucial to enhance the effectiveness of energy efficiency policies and implement measures that extend beyond mere pricing strategies. Households should also be incentivized to consume efficiently and sustainably. In addition, to avoid the rebound effect of technological progress, these countries could be advised to shift their technologies from energy efficiency to reduce ecosystem damage and raw material consumption, such as polluted air or the capture and storage of waste. However, given the importance of green technologies for the development of the CE, integrated policies are needed to reduce the rebound effect of the GRI and the CE. EU countries can limit the use of GRI resources through CE policies. They can ensure the efficient use of resources by developing and coordinating innovation policies at the national and union levels, taking into account the rebound effect of technology.

Finally, R&D investments in FI need to consider the rebound effect. The findings that FI increases the consumption of natural resources highlight the need for oversight and

supervision to mitigate the environmental impacts of FI in these countries. Accordingly, it is essential for NRM that FI companies are required to publish sustainability reports and that supervision and oversight authorities set and monitor standards for their sustainability activities. FI companies should also be encouraged to adopt and comply with energy efficiency protocols that contribute to and support environmental sustainability efforts. In this context, policymakers should provide incentives for financial innovations that improve energy efficiency while imposing additional taxes on financial technologies that lead to excessive energy consumption. In this way, companies that develop financial innovations can contribute to conserving natural resources by reducing their energy consumption. Moreover, specific requirements for FI companies to meet at least part of their energy requirements from renewable energy sources could reduce the need for energy from fossil fuels and improve NRM. Finally, providing subsidies for the development of new financial instruments, financial institutions, or innovative financial processes that contribute to the conservation of natural resources could bring out the green side of financial innovations. Other findings indicate that FD in EU countries improves NRM. Increasing the integration of financial markets, financial depth, and the development of financial institutions in the EU can support FD and contribute to natural resource conservation. Therefore, it is recommended that the European Investment Bank prioritize financial market integration as one of its strategic objectives. Moreover, the European Bank for Reconstruction and Development should allocate more resources to projects that focus on natural resource conservation and the development of innovative products.

Future limitations

This study has several limitations. First, not all EU countries could be incorporated into the analysis due to the unavailability of data for the relevant period. Additionally, certain factors that may affect NRM were excluded from this study. It is essential to emphasize the need for comprehensive findings in future research. Finally, FI, CE, and GRI, which are multidimensional concepts, can be measured via different indicators. Therefore, in future studies, various indicators of FI, CE, and GRI should be used in the analysis. Other factors that may impact NRM should be included in the research model. Additionally, other EU countries that were excluded due to data limitations should be examined for different periods. In addition, the moderating role of factors such as environmental regulation and human development in the impact of technological progress on NRM can be examined.

Moreover, broader policy implications can be drawn by adding circular ESGs to the model. In contrast, this study focuses only on natural resource conservation. Findings can be enriched with different environmental indicators. Finally, the methods used in this study also have limitations due to the period in which they were employed. Therefore, it is recommended that asymmetric alternative estimation methods be used in future studies that account for structural breaks. Thus, policy recommendations can be expanded by considering different effects.

Abbreviations

ADF Augmented Dickey–Fuller

GLS	Generalized least square
ADJ	Adjusted
GRI	Green innovation
BG	Breusch–Godfrey
HAC	Heteroscedasticity and autocorrelation consistent
BPG	Breusch–Pagan–Godfrey
IISD	International institute for sustainable development
CADF	Cross-sectionally augmented Dickey–Fuller
IMF	International monetary fund
CE	Circular economy
IRP	International resource panel
CEAP	Circular economy action plan
JB	Jarque–Bera
CER	Circular economy rebound effect
LA-VAR	Lag-augmented vector autoregressions
CIESIN	Center for international earth science information network
LM	Lagrange multiplier
CIPS	Cross-sectional im, pesaran and shin
LN	Natural logarithmic
COV	Covariance
NRM	Natural resource management
CSD	Cross section dependence
NRPI	Natural resource protection indicator
DHg	Durbin–Hausman group
OEC	Observatory of economic complexity
DHp	Durbin–Hausman panel
OECD	Organization for economic co-operation and development
DMAX	Maximum order of integration
PCA	Principal component analysis
EC	Economic complexity
PROB	Probability
ECI	Economic complexity index
R&D	Research and development
EGLS	Estimated generalized least square
REGTECH	Regulatory technology
ET	Error term
SDGs	Sustainable development goals
ETC	Et cetera
STAT	Statistics
EU	European union
UNDG	United nations sustainable development group
EUROSTAT	Statistical office of the european union
VAR	Vector autoregressions
FD	Financial development
VAR	Variance
FINTECH	Financial technology
VIF	Variance inflation factor
G7	Group of seven
WB	World bank
GDP	Gross domestic product
WWF	Worldwide fund for nature
GHG	Greenhouse gas

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